Einführung in die Rechnerarchitektur

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Überblick: Quantum Computing

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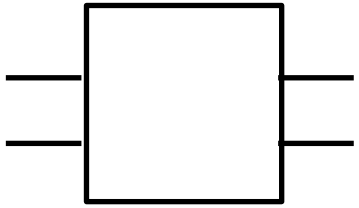
Conventional Computing

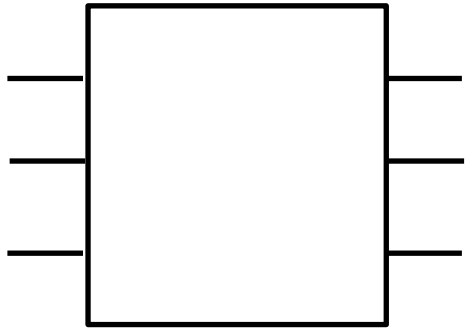
Conventional Computing

States of a System

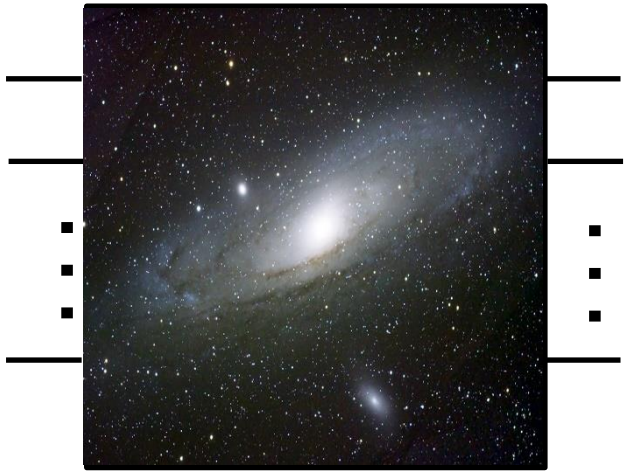
2 Inputs
→ 2^2=4 cases: 00
01
10
11



3 Inputs
→ 2^3=8 cases: 000
001
010
011
…



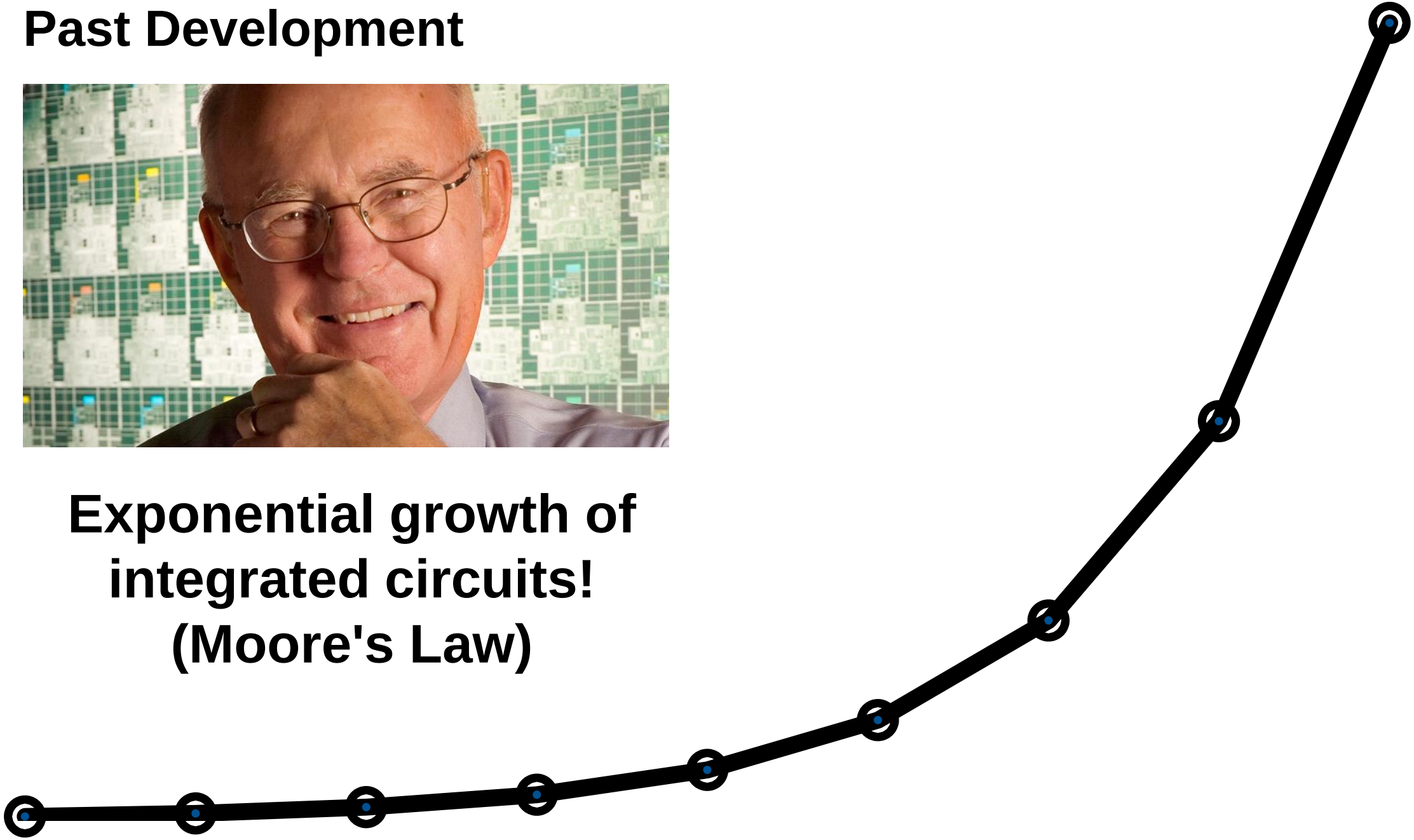
300 Inputs
→ 2^{300}=~10^{90} cases: 00000…00
00000…01
00000…10
00000…11
…



- For comparison: Number of atoms in the universe is approx. ~10^{78}

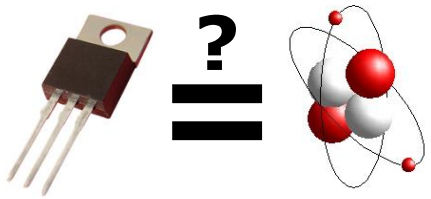
Past Development

Exponential growth of integrated circuits! (Moore's Law)

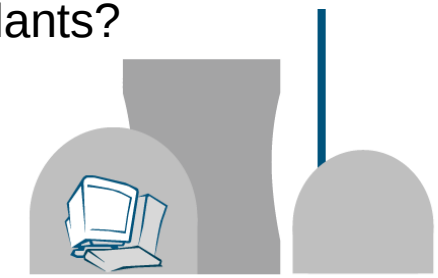


Leading to…

Transistors approaching the atomic scale?



Computer cooling as in nuclear power plants?



➔ Today‘s technologies are approaching their limits!

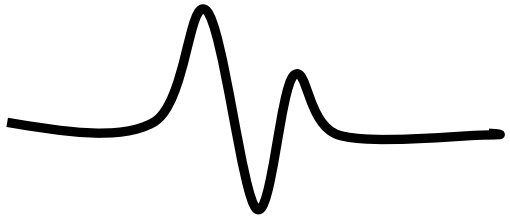
Besides that…
Many problems remain hard for today’s computing technologies

…even with further advancements.

Conventional vs. Quantum Physics

- Thus far: Newton‘s Physics
  - Describe “our world”
  - Based on particles
  - Completely deterministic

- Now: Quantum Mechanics
  - Describe the smallest „things“ in our universe
  - Described by waves
  - Completely probabilistic



  - Allows to be in more than on state at one time
  - Measurement collapses back to single state (back to “our” world)
  - **True Random** (God *does* place dice)
  - **Entanglement**
  - …

Conventional vs. Quantum Physics

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  - Describe “our world”
  - Based on particles
  - Completely deterministic

$$\mathbf{0} \qquad \mathbf{or} \qquad \mathbf{1}$$

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  - Describe the smallest „things“ in our universe
  - Described by waves
  - Completely probabilistic

$$\boldsymbol{\alpha_0} \cdot |\mathbf{0}\rangle + \boldsymbol{\alpha_1} \cdot |\mathbf{1}\rangle$$
$$\textbf{with} \quad |\boldsymbol{\alpha_0}|^2 + |\boldsymbol{\alpha_1}|^2 = \mathbf{1}$$

  - **Allows to be in more than one state at one time**
  - Measurement collapses back to single state (back to “our” world)
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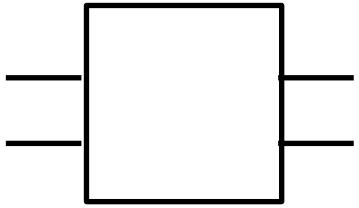

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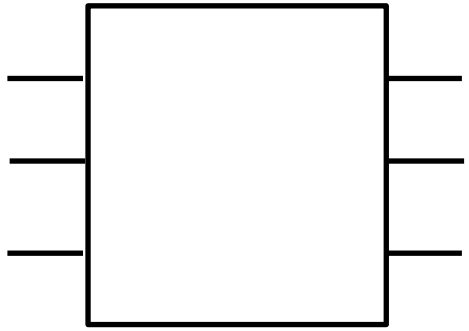

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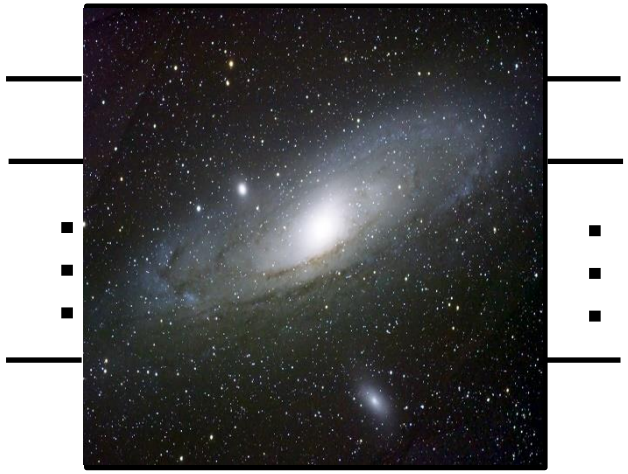

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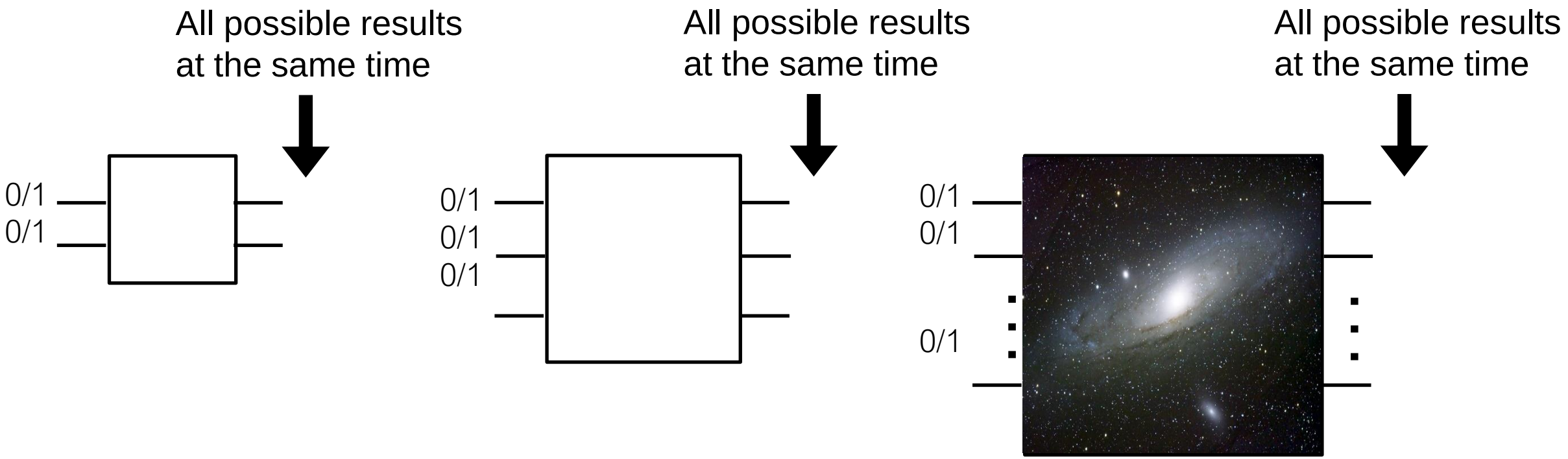
States of a System



- **But:** Measurement only gives you one possible result
- For many applications (not all!), “tricks” exist to increase probability of the desired result (e.g., through amplitude amplification, …)

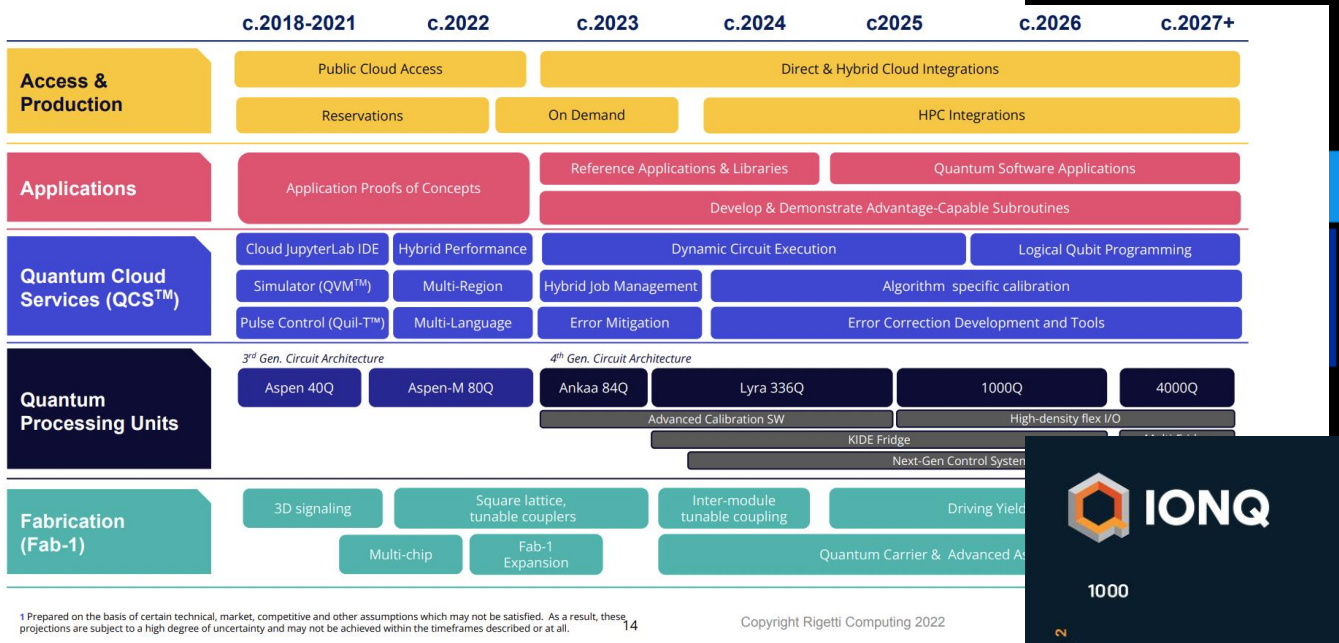
Quantum Computing: The Next Big Thing

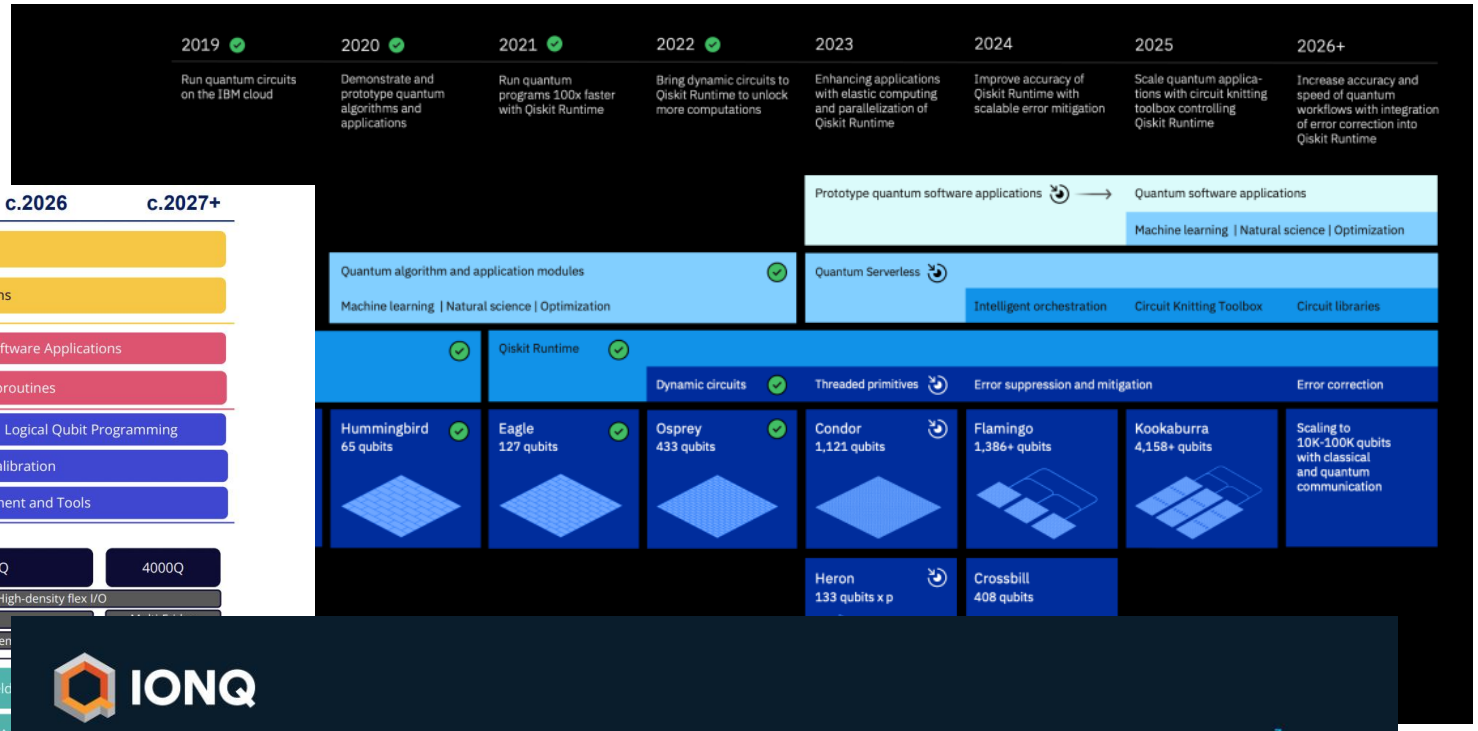
- Huge interest…
  - from big Players: IBM, Google, Microsoft, Amazon, …
  - from Startups: AQT, Rigetti, IonQ, …

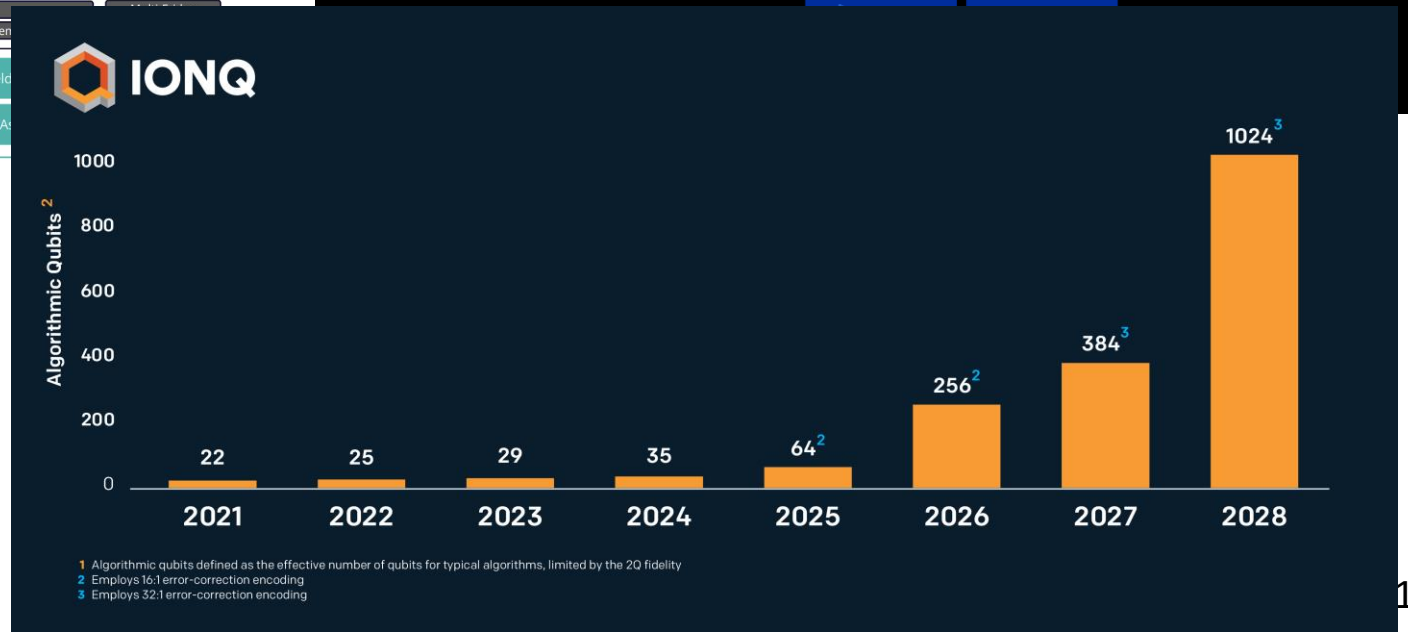
- Exponential improvements in the best case
  - Integer factoring in polynomial time
  - Break current internet cryptography

- Various further application domains
  - Nature
  - Optimization
  - Finance
  - Machine Learning
  - …

Quantum Computing: Roadmaps







Applications of Quantum Computing

Illustrative

Horizons: Primary value pools

Economic value: Incremental, Significant, Disruptive

Economic value ($ trillions)

Industry	Key segment for QC	Impact of QC[1] ~2025-30	Impact of QC[1] ~2030-35	Industry size
Global energy and Materials	Oil & gas	Incremental	Significant	5-10
	Sustainable energy	Significant	Disruptive	1-5
	Chemicals	Significant	Disruptive	1-5
Pharmaceuticals and medical products	Pharmaceuticals	Significant	Disruptive	1-5
Advanced industries	Automotive and assembly	Incremental	Significant	1-5
	Aerospace and defense	Incremental	Significant	<1
	Advanced electronics	Incremental	Significant	<1
	Semiconductors	Incremental	Significant	<1
Financial industry[1]		Incremental	Significant	>10
Telecommunications, media, and technology	Telecommunications	Incremental	Significant	1-5
	Media	Incremental	Incremental	1-5
Travel, transport, and logistics	Logistics	Incremental	Significant	5-10
Insurance		Incremental	Incremental	5-10

1. Relative impact on the industry; absolute impact depends on relative impact as well as the size of the industry.
2. Includes asset management.

Outlook

- Impact from QC is expected to be most disruptive for the chemicals and pharmaceutical industries, as quantum-computing-based simulation of molecular processes may replace the need for lab-based testing
- In the automotive industry, this may stimulate breakthroughs in battery development and new fuels
- Impact on the financial industry is more incremental; yet value at stake is high, especially in asset management

Source: McKinsey (https://www.mckinsey.de/~/media/mckinsey/locations/europe%20and%20middle%20east/deutschland/news/presse/2021/2021-09-22%20quantum%20technology%20monitor/mckinsey_quantum_technology_monitor_2021_vf.pdf)

Entwurf von Quantenanwendungen

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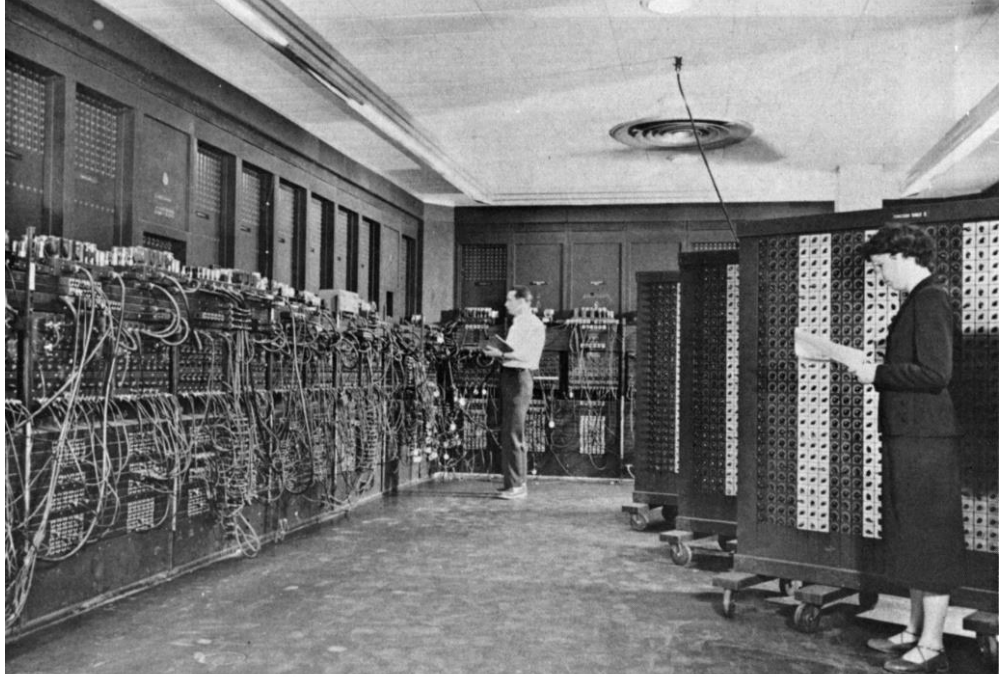
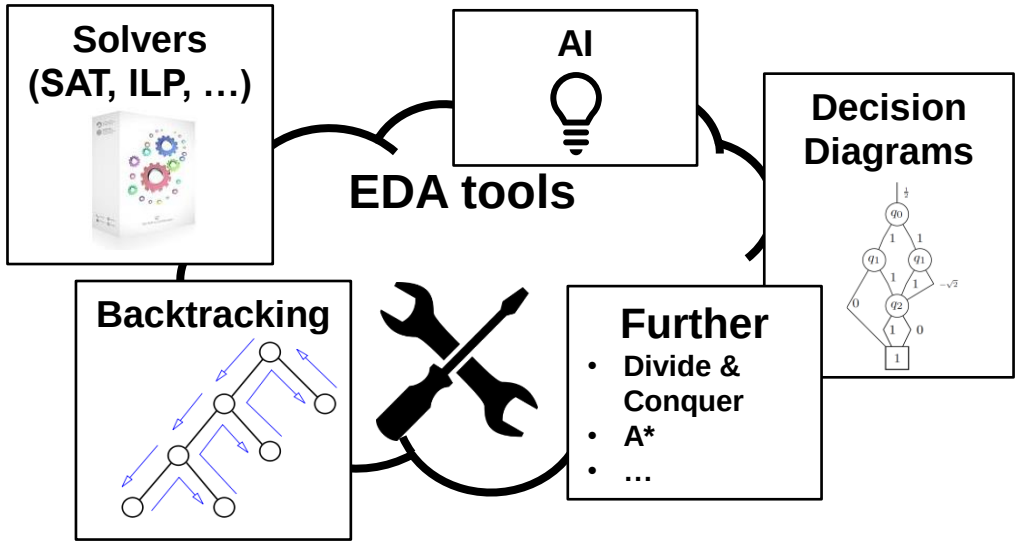
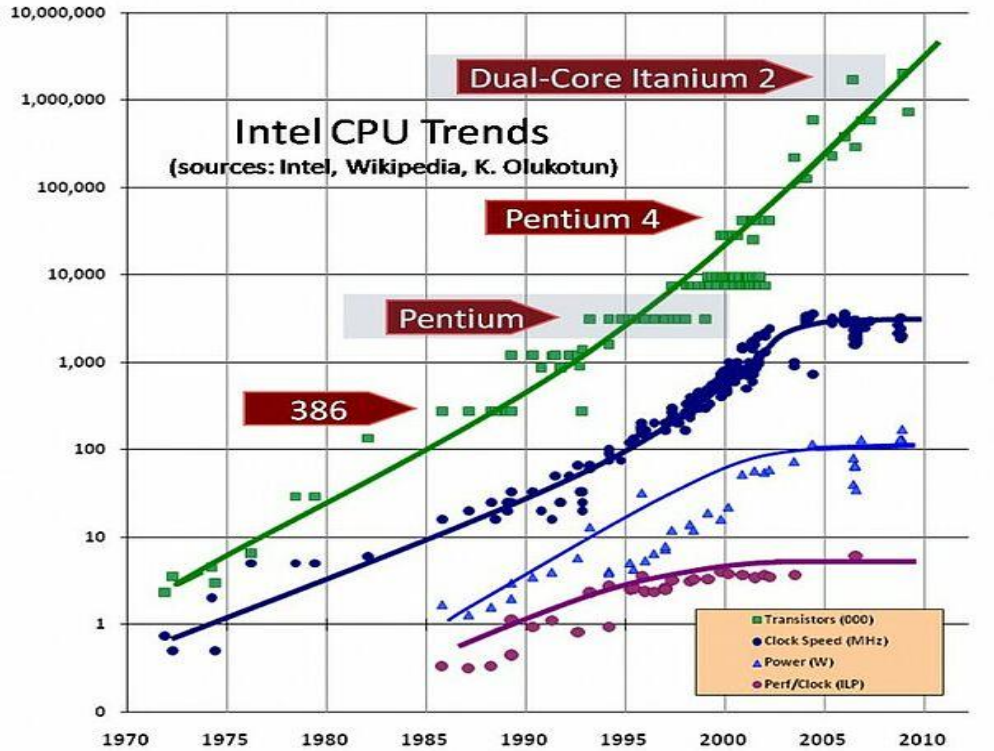
Analogy to Conventional Computers

- Similar picture if we look back in time
  - First, bulky computers
  - Moores law
  - Digital revolution



- **EDA/Software** tools enabled their success
  - **Handle** problems with **enormous complexity**





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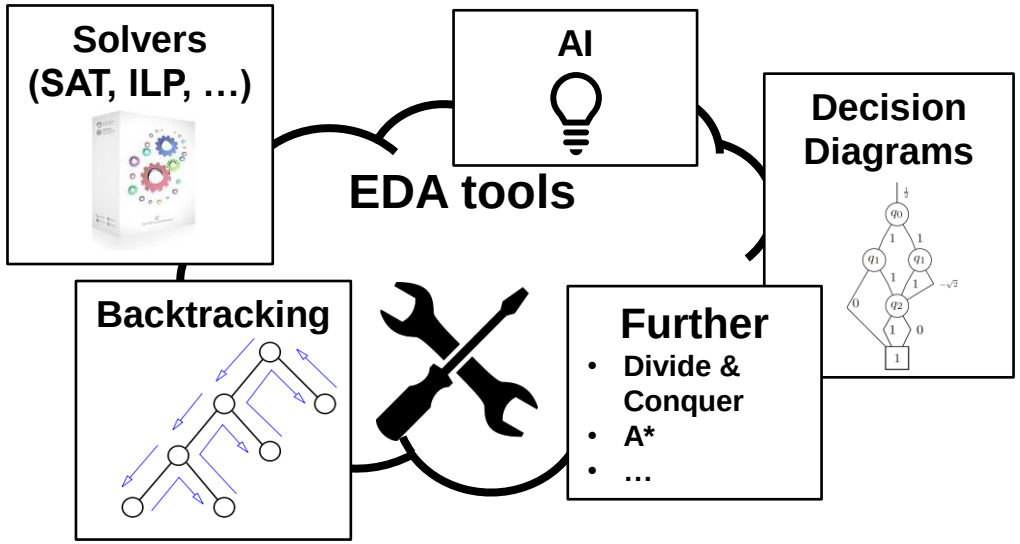
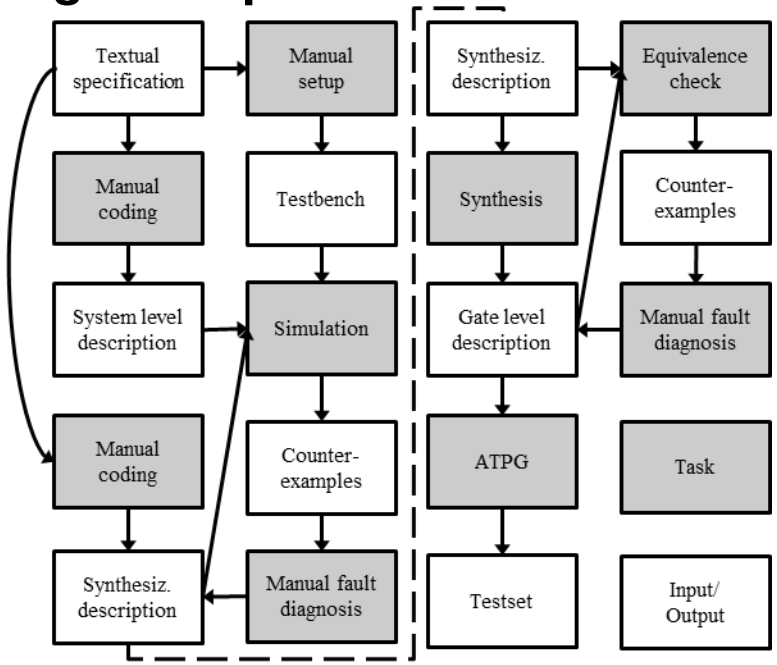


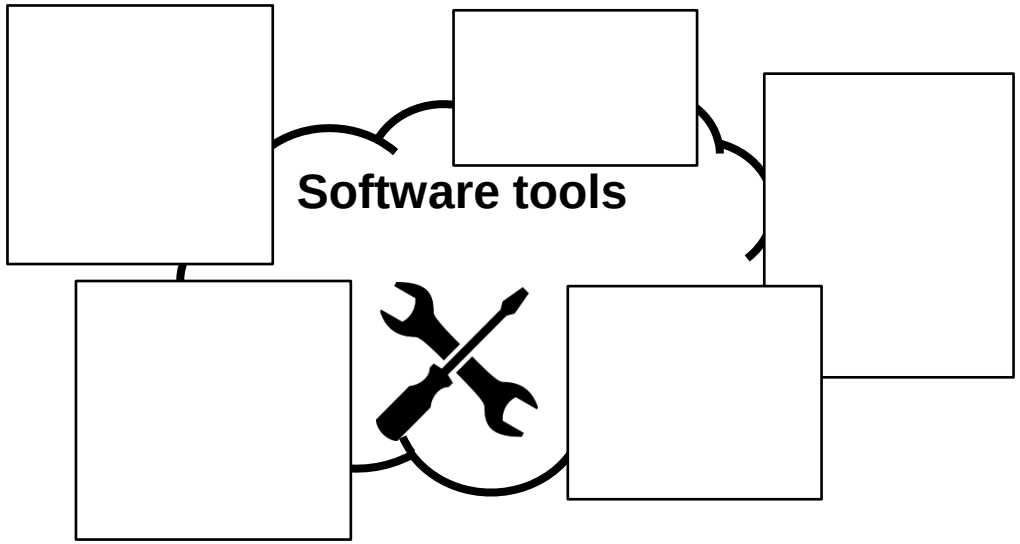
- **Emerge of powerful design/compilation flows**



EDA/Software Tools for Quantum Computers

We may end up in a situation where we have quantum computers but no efficient methods to use or design them

- **Software** tools



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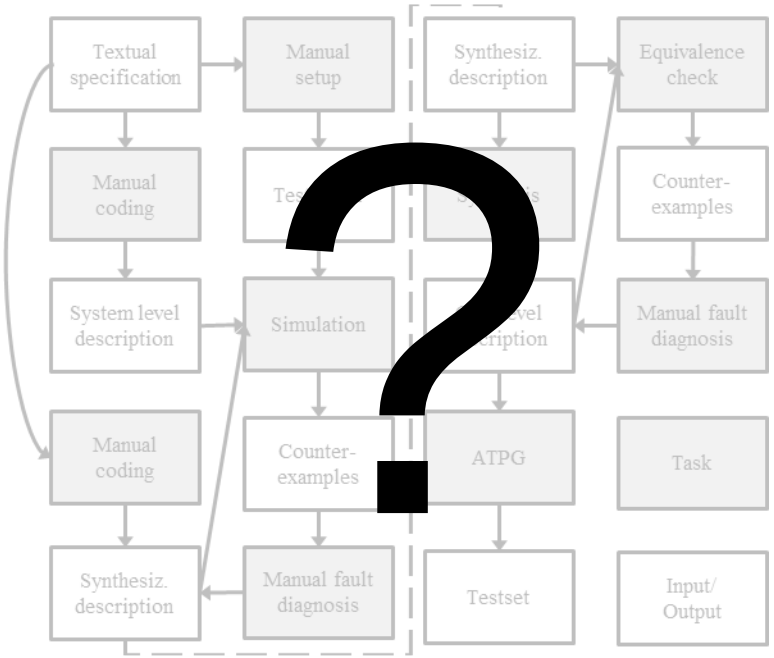
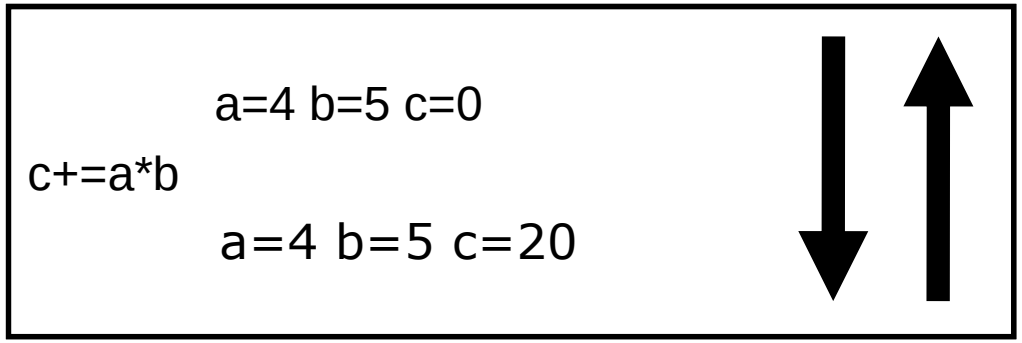


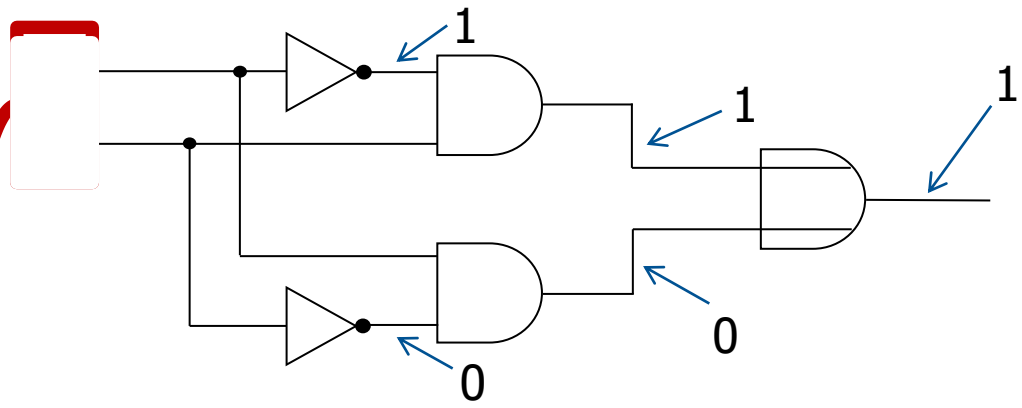
Challenges (Examples)

- "Programming" (different paradigms)
- Every quantum operation is inherently reversible



- Not to mention
  - Dealing with "0 and 1 at the same time"
  - Understanding/Handling quantum operations
  - Physical constraints
  - Error correction
  - …

- Simulation



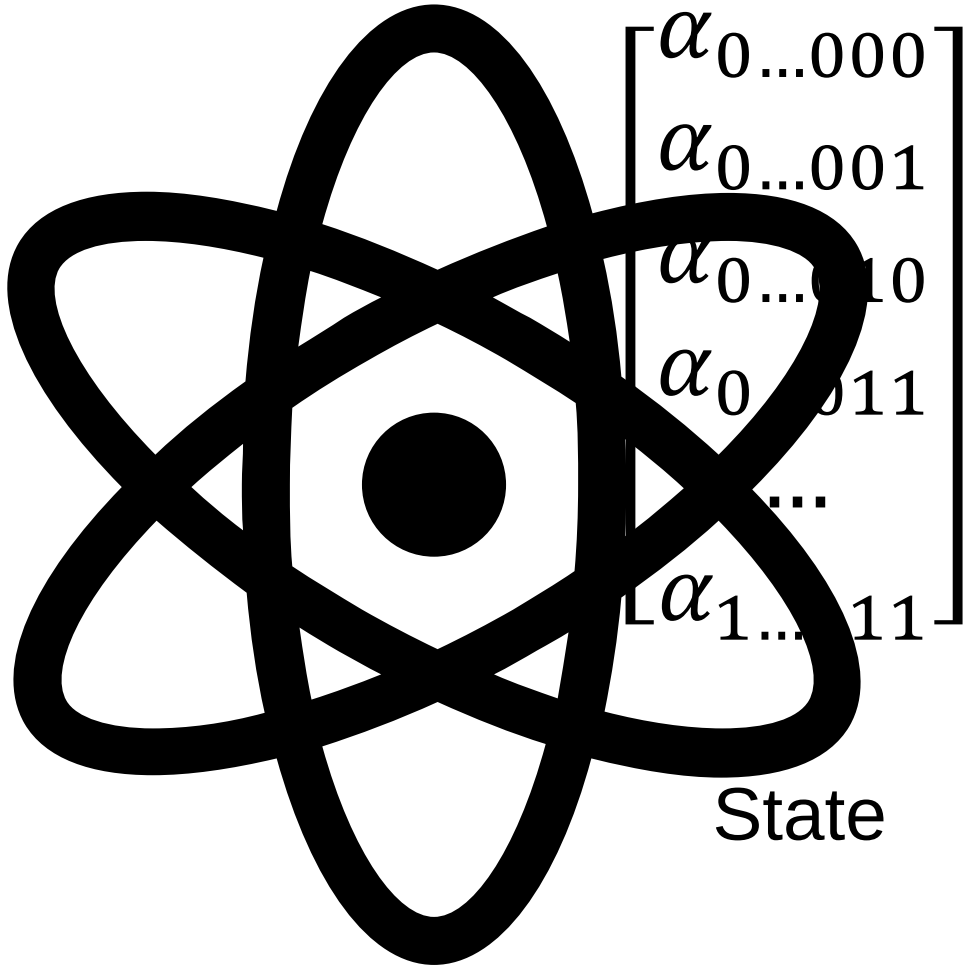
$$|\psi\rangle = \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix}$$

Exponential complexity

Already simple tasks are substantially harder for quantum computing than for conventional circuits/systems

- Further:
  - Interdisciplinarity
  - Terminology and Formalizations
  - …

Basis of Quantum Computing



$$\begin{bmatrix} \alpha_{0\ldots000} \\ \alpha_{0\ldots001} \\ \alpha_{0\ldots010} \\ \alpha_{0\ldots011} \\ \ldots \\ \alpha_{1\ldots111} \end{bmatrix}$$

State

Quantum Circuit Simulation

$$\begin{bmatrix} m_{11} & m_{12} & \dots & m_{12^n} \\ m_{21} & m_{22} & \dots & m_{22^n} \\ m_{31} & m_{32} & \dots & m_{32^n} \\ m_{41} & m_{42} & \dots & m_{42^n} \\ \dots & \dots & \dots & \dots \\ m_{2^n 1} & m_{2^n 2} & \dots & m_{2^n 2^n} \end{bmatrix} \begin{bmatrix} \alpha_{0\dots000} \\ \alpha_{0\dots001} \\ \alpha_{0\dots010} \\ \alpha_{0\dots011} \\ \dots \\ \alpha_{1\dots111} \end{bmatrix}$$

Operation

Input State

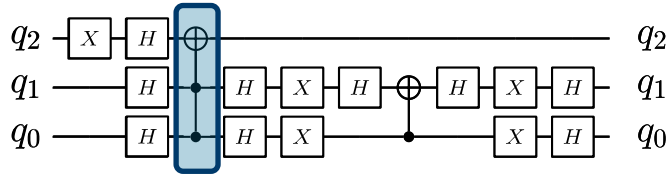
Quantum Circuit Compilation/Synthesis

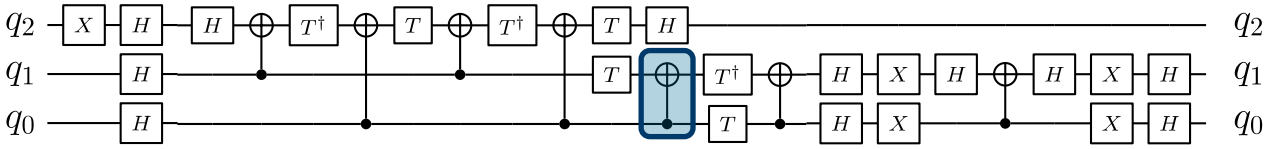
$$\begin{bmatrix} m_{11} & m_{12} & \dots & m_{12^n} \\ m_{21} & m_{22} & \dots & m_{22^n} \\ m_{31} & m_{32} & \dots & m_{32^n} \\ m_{41} & m_{42} & \dots & m_{42^n} \\ \dots & \dots & \dots & \dots \\ m_{2^n 1} & m_{2^n 2} & \dots & m_{2^n 2^n} \end{bmatrix}$$

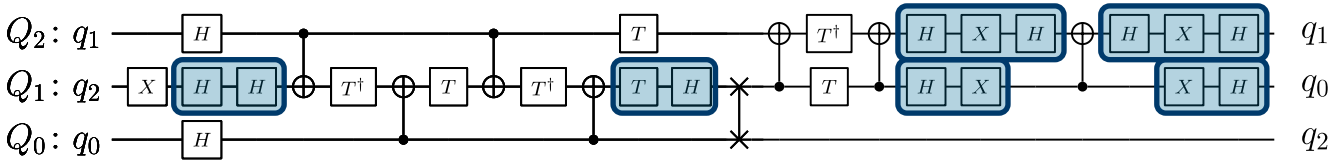
Operation/Application

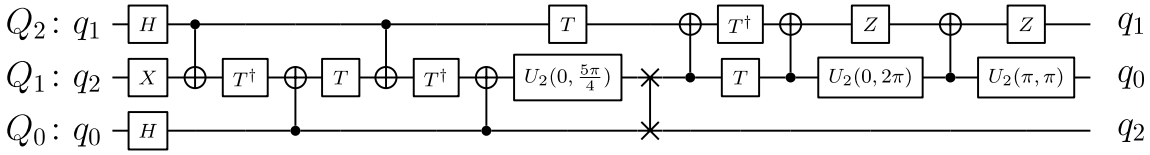
Quantum Circuit Compilation/Synthesis

$$\begin{bmatrix} m_{11} & m_{12} & \dots & m_{12^n} \\ m_{21} & m_{22} & \dots & m_{22^n} \\ m_{31} & m_{32} & \dots & m_{32^n} \\ m_{41} & m_{42} & \dots & m_{42^n} \\ \dots & \dots & \dots & \dots \\ m_{2^n 1} & m_{2^n 2} & \dots & m_{2^n 2^n} \end{bmatrix}$$





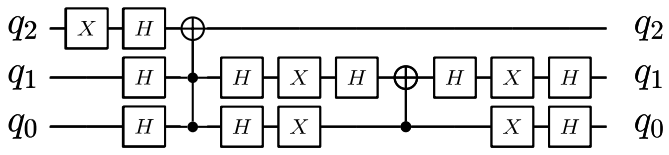




Operation/Application

Quantum Circuit Verification

$$\begin{bmatrix} m_{11} & m_{12} & \dots & m_{12^n} \\ m_{21} & m_{22} & \dots & m_{22^n} \\ m_{31} & m_{32} & \dots & m_{32^n} \\ m_{41} & m_{42} & \dots & m_{42^n} \\ \dots & \dots & \dots & \dots \\ m_{2^n1} & m_{2^n2} & \dots & m_{2^n2^n} \end{bmatrix}$$



$$\stackrel{?}{=}$$

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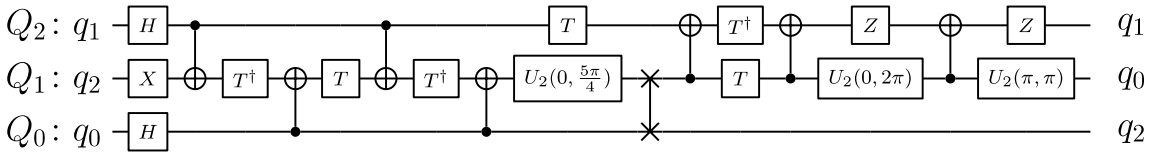


The Basis of Software for Quantum Computing…

$$\begin{bmatrix} m_{11} & m_{12} & \dots & m_{12^n} \\ m_{21} & m_{22} & \dots & m_{22^n} \\ m_{31} & m_{32} & \dots & m_{32^n} \\ m_{41} & m_{42} & \dots & m_{42^n} \\ \dots & \dots & \dots & \dots \\ m_{2^n 1} & m_{2^n 2} & \dots & m_{2^n 2^n} \end{bmatrix}$$

$$\begin{bmatrix} \alpha_{0\dots000} \\ \alpha_{0\dots001} \\ \alpha_{0\dots010} \\ \alpha_{0\dots011} \\ \dots \\ \alpha_{1\dots111} \end{bmatrix}$$

- …are vectors and matrices!

- Can be represented by 1- or 2-dimensional arrays

➔ Exponential Complexity

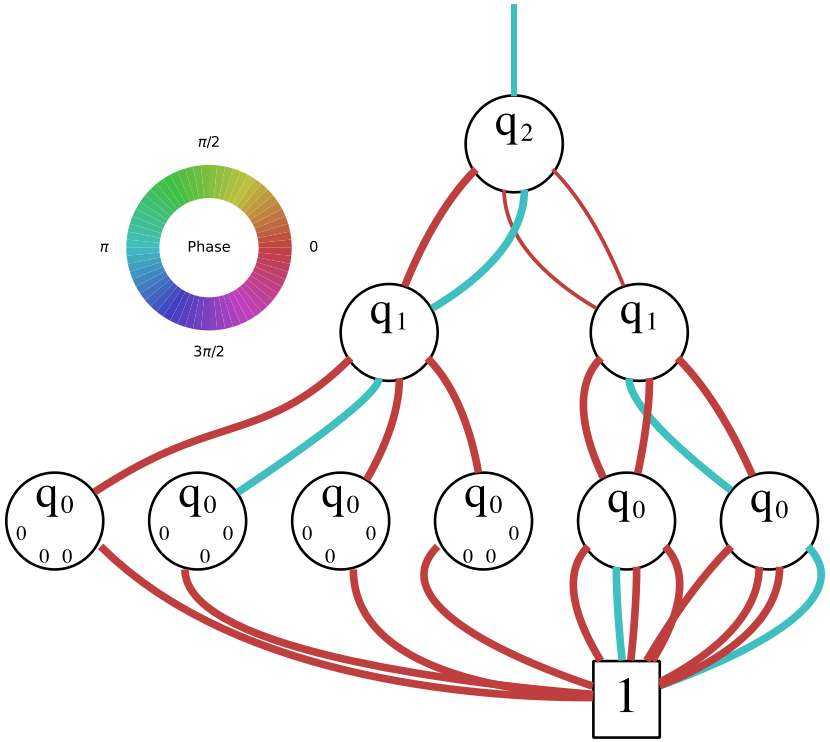
- Alternatives:
  - Decision Diagrams
  - Tensor Networks
  - ZX-Calculus

Data-structures #1

Functionality

$$-\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 0 & 0 & -1 & 1/2 & -1/2 & -1/2 & -1/2 \\ 0 & 1 & 0 & 0 & 1/2 & 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 & -1/2 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 & 1/2 & 1/2 & 1/2 & -1/2 \\ 0 & 0 & 0 & 1 & 1/2 & -1/2 & -1/2 & -1/2 \\ 0 & -1 & 0 & 0 & 1/2 & 1/2 & -1/2 & 1/2 \\ 0 & 0 & -1 & 0 & 1/2 & -1/2 & 1/2 & 1/2 \\ -1 & 0 & 0 & 0 & 1/2 & 1/2 & 1/2 & -1/2 \end{pmatrix}$$

Decision Diagram



Eventually allows for compact representation and efficient manipulation in many cases

Compilation

Conceptional algorithm

Limited Gate Set

Synthesis

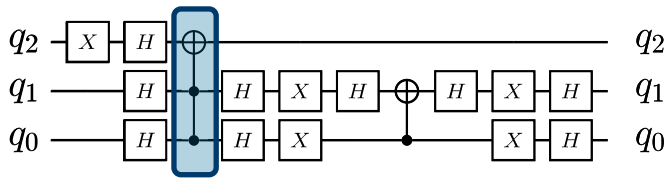
Limited Connectivity

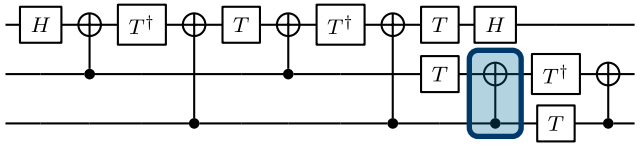
Mapping

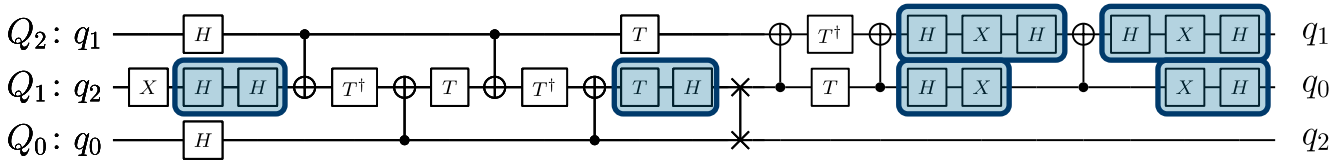
Limited Fidelity and Coherence

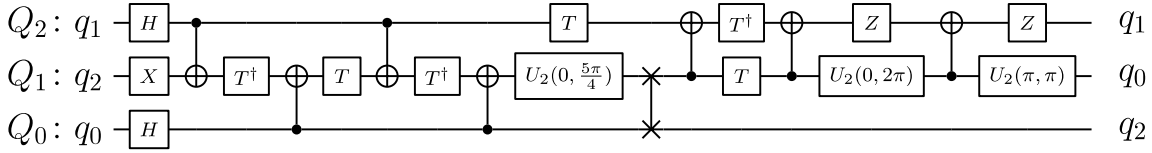
Optimizations

Actual Realization

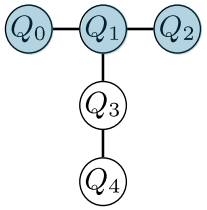








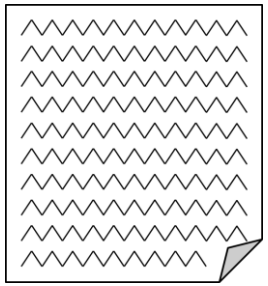
Target Device



➔ **Kind of "typical" EDA problems** (some of them proven NP-hard)

Formal Verification

Quantum Algorithm



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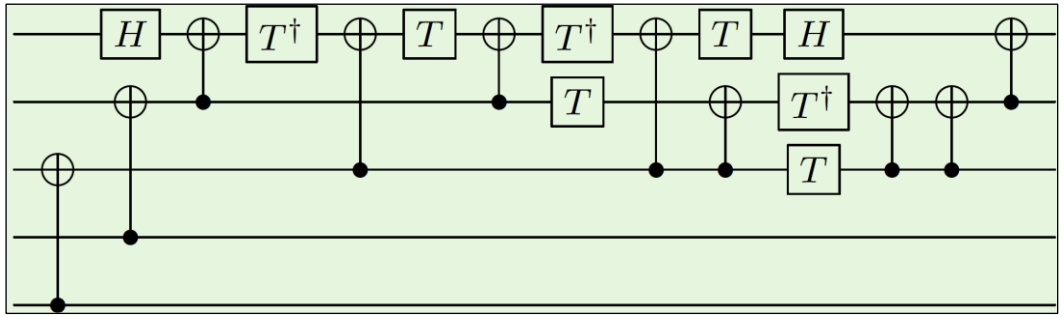

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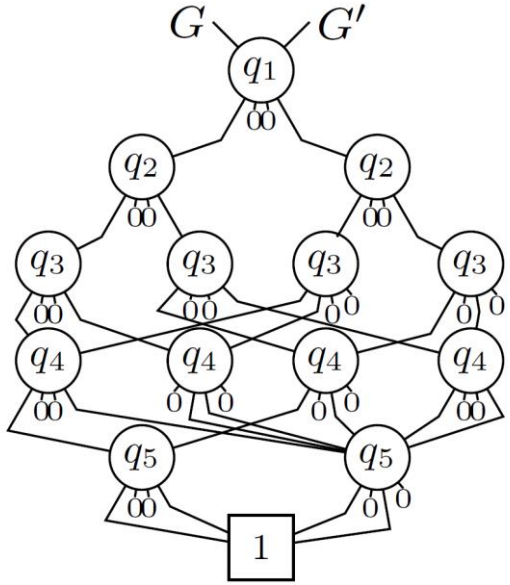

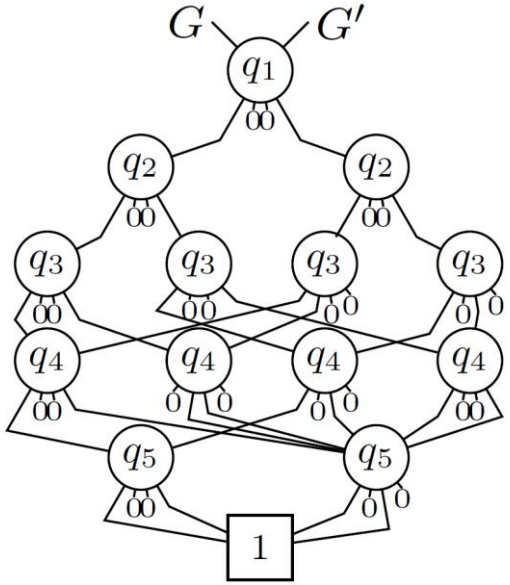

Entwurf von Quantenanwendungen

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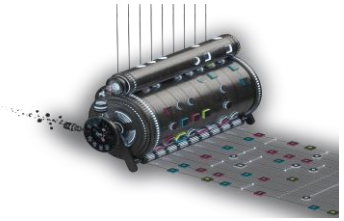
Overview: Software Tooling for Quantum

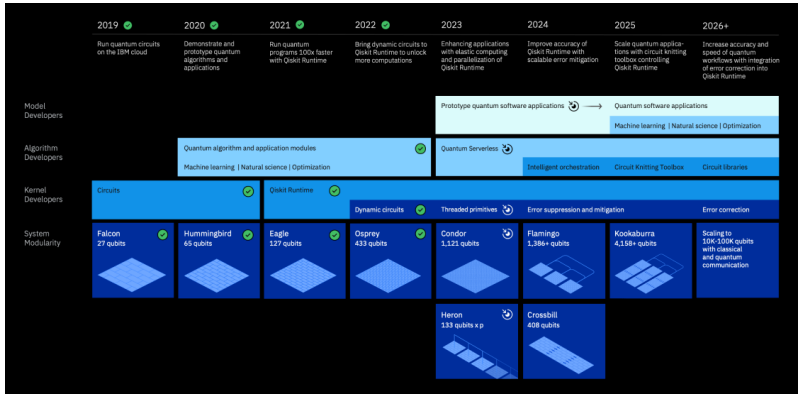
- Programming Language: Python
- Different categories of SDK providers
  - Quantum SDK and Quantum Computers (e.g., IBM/Qiskit, Google/Cirq, Quantinuum/Pytket, Xanadu/Pennylane, …)
  - Quantum SDK and platform (e.g., Microsoft Azure, AWS Braket, …)
  - Only Quantum SDK (e.g., MQT, Classiq, …)
- Functionalities of SDKs similar with differences in the details, such as syntax, supported simulators/quantum computers, …
- Simulator vs. actual Quantum Computer
- Some have special foci:
  - Pennylane: Quantum Machine Learning
  - Classiq: High-Level Software Generation
  - …



IBM Quantum

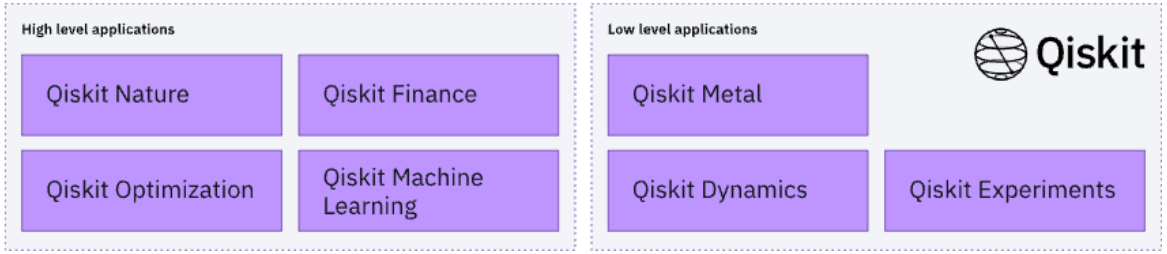
- One of the largest Quantum SDKs: Qiskit
- Access provider to all (currently 24) IBM Quantum Computers, e.g.:

Name	Qubits	QV	CLOPS	Status
ibm_washington	127	64	850	Online
ibm_sherbrooke	127	32	904	Online
ibm_kyiv Exploratory	127	-	-	Online
ibm_ithaca Exploratory	65	-	-	Online



- Pricing: ~1,6 € / Qiskit Runtime Seconds
- Huge community effort:

Qiskit Textbook:
https://qiskit.org/learn/



High- and Low-Level Tutorials:
https://qiskit.org/documentation/

Youtube Channel:
https://www.youtube.com/qiskit

The Munich Quantum Toolkit (MQT)

MUNICH QUANTUM TOOLKIT

https://www.cda.cit.tum.de/research/quantum/mqt/

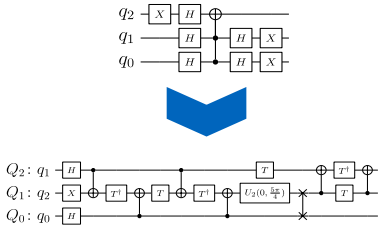
The Munich Quantum Toolkit (MQT)

Application

- Classical problem encoding
- Quantum solution execution
- Solution Decoding

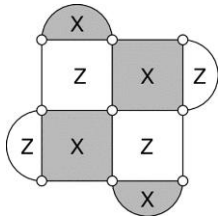
Compilation

- Best compilation options
- Reversible synthesis
- Technology-specific Mapping
- Qudit Compilation



Error Correction

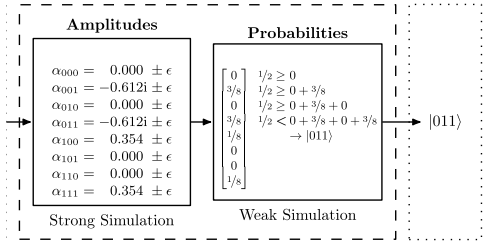
- Decoding algorithms
- Automated code construction and numerical simulations



Application → Simulation → Compilation → Verification → Error Correction → Hardware

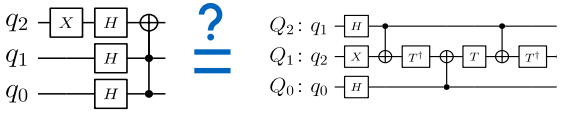
Simulation

- Decision Diagram based simulation of gate-based quantum circuits
- Strong & weak simulation
- Noise-aware simulation



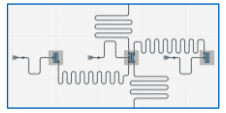
Verification

- Equivalence checking of quantum circuits



Hardware

- Application specific physical design for superconducting platform



Data Structures & Core Methods

- Efficient data structures
- Dedicated core methods (optimal and heuristic)
- Based on C++ and Python

Decision Diagrams

Tensor Networks

ZX-Calculus

SAT/SMT Solvers

Machine Learning

Heuristics

Check it out!

https://www.cda.cit.tum.de/research/quantum/mqt/